

Supplementary Information for

FEWT: Frequency-Enhanced Wavelet-based Transformer for Multimodal Wheeled Bimanual Manipulation

Supplementary material

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A. Smart Tactile Fabric

This study develops a customized flexible tactile sensing module (Low-cost, the fabrication cost is only \$4), termed Smart Tactile Fabric (STF), as illustrated in **Fig. S1**.

Structurally, the tactile fabric integrates a 15×15 sensing array, providing high spatial resolution. Its core sensing mechanism relies on an internal piezoresistive functional layer. Upon the application of external physical stress, the thin film exhibits a significant variation in electrical resistance, thereby precisely transducing subtle mechanical pressures into measurable electrical signals. The underlying transduction principle operates as follows: external loads applied to the fabric surface induce localized deformation within the pressure-sensitive layer, which subsequently alters the resistance of the corresponding sensing units. These resistance variations are captured by interwoven conductive yarn electrodes and converted into analog voltage signals. Subsequently, the analog signals are conditioned by a custom-designed signal processing board, followed by analog-to-digital (A/D) conversion and data packaging via an Arduino Nano microcontroller.

Finally, the digitized data is transmitted to a host computer for real-time decoding and visualization of the tactile information stream, with the output value of each sensing unit scaled to a range of 0 to 255. The relevant code of STF is now open source: https://github.com/Smart-Tactile-Fabric/STF_touch_visualization.

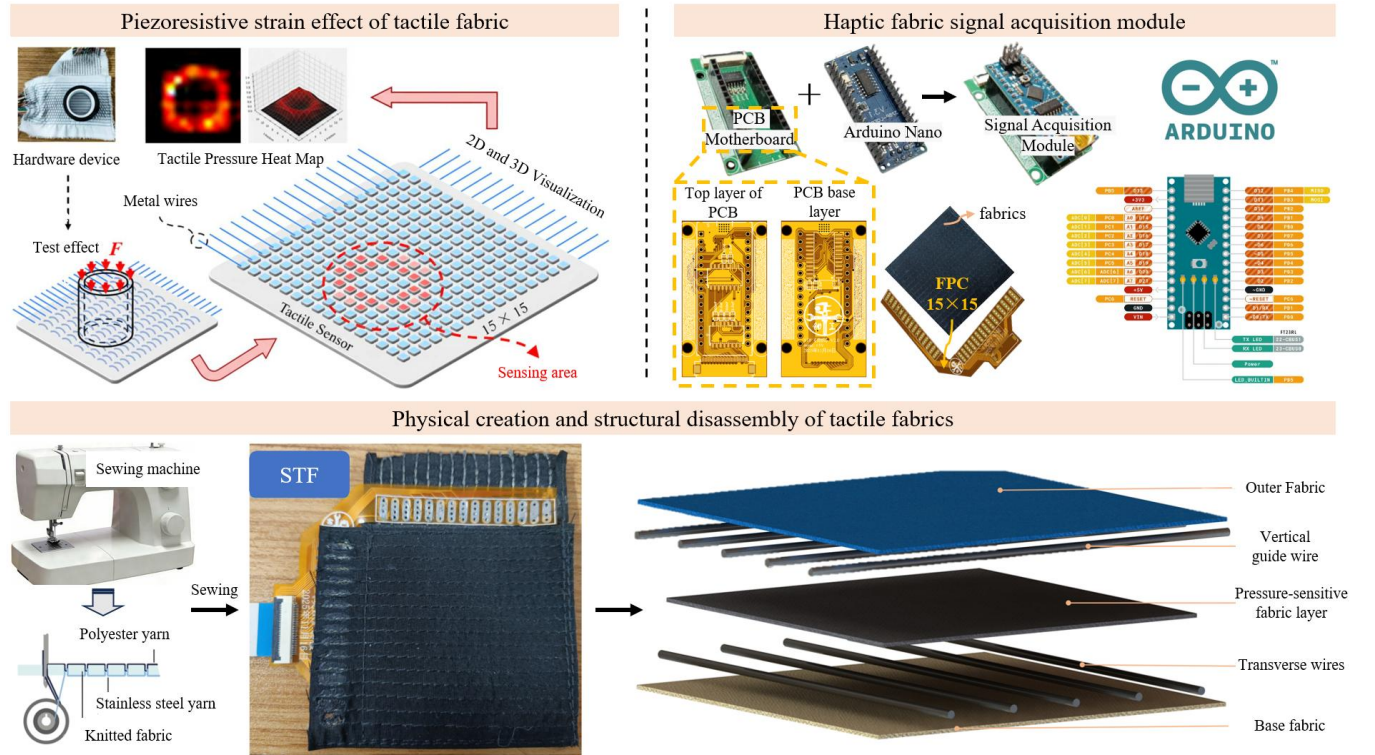


Fig. S1: Robot tactile perception module (Smart Tactile Fabric, STF).

B. Comparison of Parameters and FLOPs

Table S1 shows that, based on the ACT Policy baseline and Backbone (ResNet18), the FE-EMA module reduces the computational complexity (FLOPs) compared to the EMA module. Where the Simulation Tasks (ACT baseline) are **Cube Transfer** and **Bimanual Insertion**, respectively.

FLOPs were profiled using THOP with RGB image of $3 \times 480 \times 640$, a 14-dimensional joint-state input, and an action-chunk length of $k = 100$. The results were normalized to one sample, and one multiply-accumulate operation was counted as two floating-point operations.

Table S1. Comparison of profiled model parameters and per-sample forward-pass FLOPs under the ACT architecture.

Method	Backbone	Simulation Tasks			
		#.Param.	FLOPs	Δ Params vs. ACT	Δ FLOPs vs. ACT
Baseline (ACT)	ResNet18	60.7566 M	37570.51 M	–	–
EMA + ACT		60.7592 M	37616.59 M	+0.0043%	+0.1226%
FE-EMA + ACT (ours)		60.7621 M	37596.91 M	+0.0091%	+0.0703%

As shown in Table S1, the proposed FE-EMA module introduces only a marginal computational overhead under the ACT–ResNet18 architecture. Compared with the ACT baseline, the profiled parameter count increases from 60.7566 M to 60.7621 M, corresponding to an increase of only 0.0055 M, or approximately 0.0091%. Meanwhile, the computational cost increases from 37.57051 G to 37.59691 G FLOPs, representing an overhead of only 26.40 M FLOPs, or approximately 0.0703%. These results indicate that the frequency-enhancement mechanism does not rely on substantial model expansion.

Compared with conventional EMA, FE-EMA contains approximately 0.0029 M additional profiled parameters but requires 19.68 M fewer FLOPs, corresponding to a reduction of approximately 0.0523% in total forward-pass computation. This seemingly different trend between parameter count and FLOPs arises from the architectural design of FE-EMA: the wavelet decomposition first separates and down samples the multi-frequency feature components, after which the main convolutional enhancement operations are performed on lower-dimensional feature representations. Consequently, FE-EMA can introduce lightweight frequency-fusion parameters while avoiding a proportional increase in computational cost.

Table S2 shows that, based on the Diffusion Policy (DDIM) baseline and Backbone (ResNet18), the FE-EMA module reduces the computational complexity (FLOPs) compared to the EMA module. The DDIM policy uses a ResNet18 visual backbone, a U-Net noise predictor, a chunk size of 32, and 10 denoising steps during testing. Parameters and FLOPs were calculated using the same profiling convention for all three configurations. FLOPs represent computational complexity under the specified profiling implementation and should not be interpreted as measured wall-clock inference latency.

Table S2. Comparison of profiled model parameters and per-sample forward-pass FLOPs under the Diffusion Policy architecture.

Method	#.Param.	FLOPs	Δ Params vs. DDIM	Δ FLOPs vs. DDIM
Baseline (DDIM)	101721.46 K	339266.27 M	–	–
EMA + DDIM	101724.05 K	339957.52 M	+0.0025%	+0.2037%
FE-EMA + DDIM (ours)	101726.95 K	339662.33 M	+0.0054%	+0.1167%

Table S2 further evaluates the computational portability of FE-EMA under the substantially more computationally demanding Diffusion Policy architecture. The DDIM baseline contains 101721.46 K profiled parameters and requires 339266.27 M FLOPs. Incorporating FE-EMA increases the parameter count by only 5.49 K, corresponding to a negligible relative overhead of approximately 0.0054%. The corresponding computational increase is 396.06 M FLOPs, equivalent to only 0.1167% of the DDIM baseline.

In comparison, conventional EMA introduces an additional 691.25 M FLOPs over the DDIM baseline, whereas FE-EMA introduces only 396.06 M FLOPs. Therefore, FE-EMA reduces the incremental computational cost introduced by the attention module by approximately 42.70%. Although FE-EMA contains approximately 2.90 K more profiled parameters than conventional EMA, it requires 295.19 M fewer FLOPs, corresponding to a reduction of approximately 0.0868% in the total forward-pass computation.

More importantly, when the computational results in **Table S1** are considered jointly with the ablation results in **Table 2**, the performance improvements cannot be attributed to brute-force model scaling. With only 0.0091% additional parameters and 0.0703% additional FLOPs over ACT, FE-EMA improves the final Transfer-stage success rate in Cube Transfer from 86%

to 100% and the highly constrained Insert-stage success rate in Bimanual Insertion from 32% to 58%, corresponding to absolute improvements of 14 and 26 percentage points, respectively. Compared with conventional EMA, FE-EMA achieves an 18-percentage-point improvement in the Insert phase while requiring fewer FLOPs. Therefore, FE-EMA provides a favorable performance–computation trade-off, particularly in contact-rich manipulation phases that demand fine-grained spatial and frequency-domain representations.

C. Training Details

We adopt the hyperparameters reported in the ACT Policy paper, with the sole modification of reducing the batch size from 8 to 4. The complete hyperparameter settings are summarized in **Table S3**. We deploy our policy with inference on a desktop with an NVIDIA RTX 3060 GPU. The sensitivity curve analysis of the chunk size hyperparameter is shown in **Fig. S5**.

Table S3. Hyperparameters of the FEWT (based on ACT Policy).

Hyperparameter	FEWT
Learning rate	1e-5
Batch size	4
# encoder layers	4
# decoder layers	7
Feedforward dimension	3200
Hidden dimension	512
# heads	8
Chunk size	100
KL weights	10

For the Diffusion Policy (DDIM) baseline, we provide the corresponding experimental details, with the complete hyperparameter settings summarized in **Table S4**.

Table S4. Hyperparameters of Diffusion Policy.

Learning rate	1e-4
Batch size	4
Chunk size	32
Scheduler	DDIM
Train and test diffusion steps	50, 10
Backbone	ResNet18
Noise predictor	U-Net
image augmentation	RandomCrop(ratio=0.95)& ColorJitter(brightness=0.3,contrast=0.4,saturation=0.5)& RandomRotation(degrees=[-5.0,5.0])

For the two simulation tasks, *Cube Transfer* and *Bimanual Insertion*, the dynamic weight α is consistently maintained at approximately 0.5 during training (see **Fig. S2**). This suggests that the cross-spatial and frequency-domain feature weights are well balanced during model training. Dynamically fusing the time-domain and frequency-domain features allows effective capture of information across different scales.

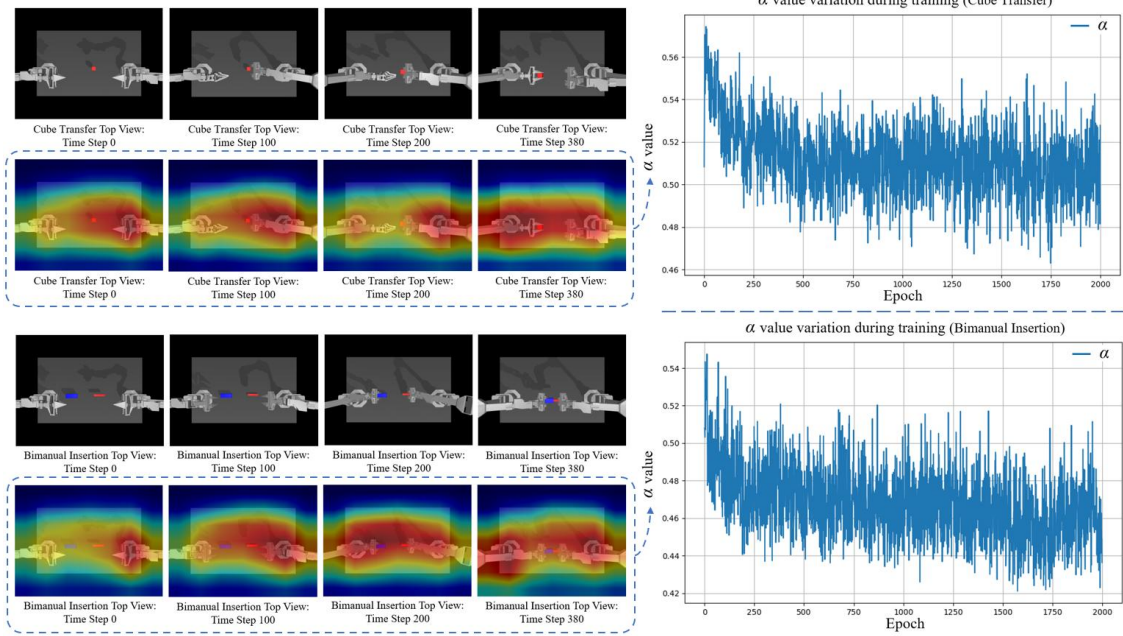


Fig. S2: Visualization of the dynamic weight α in the FE-EMA module (training phase, effect of 2000 epoch under different simulation tasks).

D. The hardware part of the robot

The construction of mobile dual-arm manipulation platform mainly consists of two 7-dof (arm + gripper) robotic arms (*ViperX-300*), a two-wheeled differential mobility chassis (*Diablo*), three RGB cameras (Logitech C922x webcams), and an inertial measurement unit (IMU sensor). The purpose of the chassis linkage restraint parts is to prevent skidding during chassis steering movements (See Fig. S3).

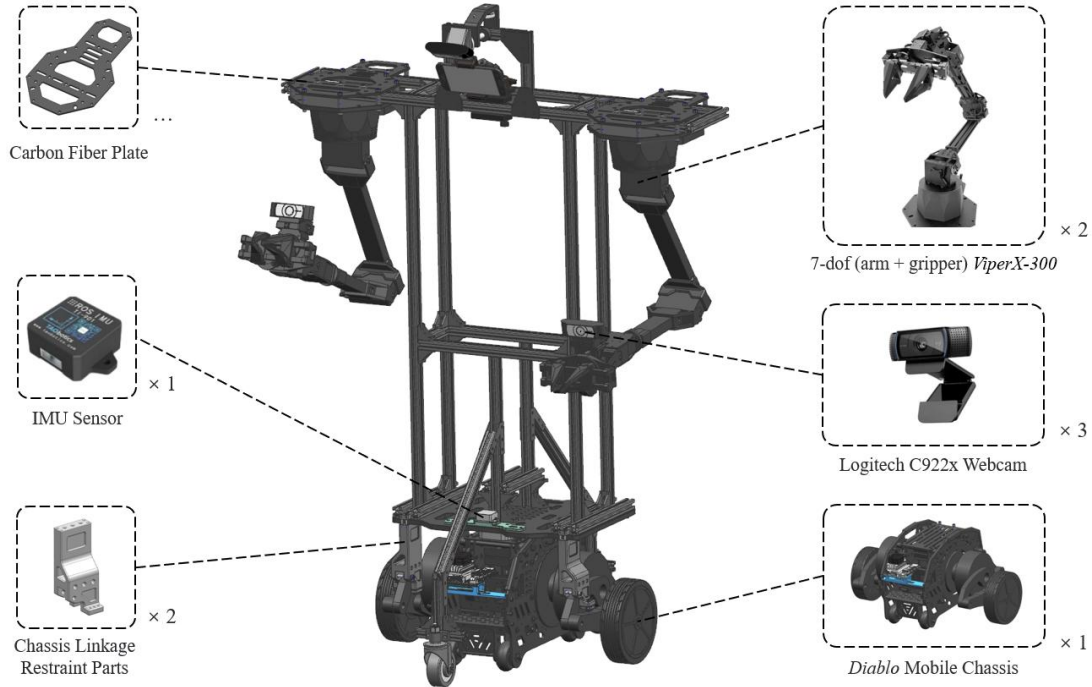


Fig. S3: The design of mobile dual-arm manipulation platform.

E. Summary of Experimental Data Details

To ensure rigorous reproducibility and to clarify the computation of partial success rates in our multi-stage tasks, we explicitly define the measurable kinematic and geometric criteria for each task phase as follows: **Reach/Approach**: The Euclidean distance between the end-effector's center and the target object's bounding box is reduced to less than 3 cm. **Grasp**: The gripper aperture is constrained to the object's resting width, and the relative position between the end-effector and the object's center of mass remains stable (variance < 0.5 cm) upon subsequent movement. **Lifted**: The vertical z-coordinate of the object's lowest point exceeds 5 cm above its initial resting surface. **Transfer**: The Euclidean distance in the x-y plane between the object's center of mass and the designated target coordinates is less than 3 cm. **Splicing**: The geometries of the two charm components are successfully aligned and connected, with the relative alignment error between their center axes being less than 0.5 cm. **Opened**: The linear displacement of the articulated target mechanism (e.g., the drawer) exceeds 10 cm from its fully closed state. **Place**: The object's vertical z-coordinate aligns with the target surface height, its linear velocity approaches zero (< 0.01 m/s), and the gripper aperture widens, causing the end-effector to cleanly disengage from the object.

For the simulated and real-world tasks, we report training with scripted data, with 50 policy evaluations each. The detailed partial success rates, strictly based on these measurable thresholds, are summarized in **Table S5**. In the MuJoCo simulation evaluations, the object initialization positions for each simulation task are randomly generated within a given interval.

Table S5. Summary of success rates at each task stage in simulation and real-world experiments.

Policy	<i>Building blocks (sim)</i>			<i>Drawer Storage (sim)</i>		
	grasp	place	place	grasp	lifted	place
DP (DDPM)	94±1.8	80±2.5	80±2.5	94±1.5	86±2.2	76±2.8
DP (DDIM)	98±1.5	88±1.8	88±1.8	96±1.3	90±1.8	84±2.5
ACT	98±1.3	94±1.5	94±1.5	98±1.3	94±1.5	94±1.8
FEWT (without IMU)	100±0.0	96±0.8	96±0.8	100±0.0	100±0.0	100±0.0
Policy	<i>Lifted Curtain (sim)</i>			<i>Strip Transfer (sim)</i>		
	lifted	grasp	place	grasp	lifted	place
DP (DDPM)	94±1.3	82±2.2	82±2.5	96±1.8	84±2.2	80±2.5
DP (DDIM)	96±1.8	90±1.5	90±1.8	98±1.3	88±1.8	86±1.8
ACT	96±1.5	84±1.8	84±1.5	98±1.5	96±1.5	96±1.5
FEWT (without IMU)	100±0.0	92±0.8	92±0.8	100±0.0	100±0.0	100±0.0
Policy	<i>Charm Splicing (real)</i>			<i>Throw Paper (real)</i>		
	lifted	approach	splicing	reach	grasp	place
DP (DDPM)	100±0.0	86±2.2	74±2.2	88±1.8	76±2.5	70±2.8
DP (DDIM)	100±0.0	94±1.5	88±1.5	94±1.5	84±1.8	80±2.2
ACT	100±0.0	96±1.5	92±1.5	96±1.5	90±1.5	86±1.8
FEWT (with IMU)	100±0.0	100±0.0	98±0.8	100±0.0	96±0.8	94±0.8
Policy	<i>Drawer Storage (real)</i>			<i>Doll Storage (real)</i>		
	reach	opened	place	reach	transfer	place
DP (DDPM)	88±2.2	76±2.5	68±2.8	88±1.8	72±2.5	60±2.8
DP (DDIM)	94±1.8	80±1.8	76±2.2	94±1.5	80±1.8	72±2.2
ACT	96±1.5	82±1.5	74±1.8	96±1.3	82±1.5	70±1.8
FEWT (with IMU)	100±0.0	92±0.8	84±0.8	100±0.0	94±0.8	82±0.8

Real-Robot System Implementation and Deployment Details:

To ensure the reproducibility of our real-world experiments and provide comprehensive insights into our hardware-and-learning framework, we detail the system-level implementation parameters below:

Teleoperation Mapping: We utilize a 1:1 Direct Joint Mapping between the exoskeleton master arm (WX-250s) and the ViperX-300 slave arm. Because both systems share isomorphic kinematics, joint angles are directly mirrored without the need for complex inverse kinematics solvers, minimizing computational delay and singularity issues.

Camera Calibration: As the FEWT framework employs an end-to-end visual imitation learning paradigm, the policy implicitly learns spatial representations from raw pixels, negating the necessity for strict intrinsic or extrinsic camera calibration. To prevent domain shift between the training and deployment phases, all cameras are rigidly mounted, and dynamic features such as auto-focus, auto-exposure, and auto-white balance are completely disabled.

Multimodal Synchronization: The system integrates asynchronous sensors, including cameras, an IMU, and arm encoders. We employ the `message_filters::ApproximateTime` synchronizer within the Robot Operating System (ROS) to fuse these distinct data streams into unified temporal observation tokens based on their closest hardware timestamps.

Inference Rate and Control Latency: The neural network policy is deployed on a local desktop computer equipped with an NVIDIA RTX 3060 GPU. The low-level controller executes buffered action commands at 50 Hz, whereas the policy inference and the end-to-end sensing-to-actuation pipeline operate asynchronously. Regarding memory efficiency, the framework requires only approximately 4~5 GB VRAM for training and approximately 1~2 GB for real-time inference. This lightweight footprint guarantees zero frame drops and enables future deployment on mobile edge devices like the NVIDIA Jetson. The overall system control latency—encompassing camera frame buffering, ROS topic transmission, and motor actuation—is measured to be between 60 and 80 ms.

Controller Interface: The entire hardware ecosystem is strictly unified under the Robot Operating System (ROS Noetic) framework. The high-level neural network policy runs on the local desktop, publishing target joint positions and chassis velocity commands as standard ROS messages. The respective hardware ROS driver nodes (for the ViperX-300 arms, Diablo chassis, and IMU) subscribe to these topics to execute low-level actuations and data acquisition. This fully ROS-integrated architecture ensures a standardized, highly decoupled, and low-latency communication pipeline.

Safety Constraints and Failure Handling: To ensure safe operation in contact-rich tasks, strict software-level safety constraints are enforced. **Action clipping** is applied to the policy's output to ensure joint angles and target velocities strictly reside within safe kinematic workspaces and hardware limits. In terms of failure handling, if a ROS communication timeout exceeds 100 ms or the policy generates anomalous (e.g., NaN) outputs, a fallback safety node automatically publishes zero-velocity commands to halt all robot movements. Furthermore, an independent hardware Emergency Stop (E-stop) switch is continuously monitored by a human supervisor during all real-world autonomous inferences.

To explicitly isolate and quantify the independent contribution of the Inertial Measurement Unit (IMU) in dynamic tasks, we conducted a targeted ablation study focusing on real-world mobile manipulation scenarios (*Object Storage* and *Bimanual Carry*). Because these tasks require coordinated whole-body control involving a two-wheeled differential chassis, overcoming base instability and dynamic perturbations is highly challenging. As presented in **Table S6**, we evaluated three diverse policy architectures: DDIM, ACT, and our proposed FEWT. To ensure rigorous variable isolation, the tactile perception modality (+ STF) was uniformly applied across all trials, isolating the IMU input as the sole independent variable.

Table S6. Ablation experiment on IMU integration for real-world robotic applications.

Method + STF (Tactile)	<i>Object Storage</i> (Mobile)	<i>Bimanual Carry</i> (Mobile)	Overall Success Rate
DDIM (without IMU)	48±1.8	52±1.5	49.8±1.5
DDIM (with IMU)	56±1.5	58±1.8	57.2±1.5
ACT (without IMU)	52±1.5	56±1.3	54.2±0.8
ACT (with IMU)	58±1.5	62±1.5	60.3±1.4
FEWT (without IMU)	90±1.3	86±1.5	88.2±0.8
FEWT (with IMU)	98±0.8	98±0.8	98.3±0.8

The quantitative results clearly demonstrate the independent efficacy of the IMU. Across all evaluated methods, the integration of IMU data consistently yields an increase in the overall success rate. The physical rationale is that the IMU provides continuous, high-frequency linear acceleration and angular velocity feedback, allowing the closed-loop policy to adaptively compensate for the low-frequency drift and high-frequency vibrations inherent in wheeled locomotion. Furthermore, the FEWT model achieves the highest overall success rate of 98.3% when equipped with the IMU. This significant performance advantage indicates that the Time-Series Discrete Wavelet Transform (TS-DWT) module within the FEWT framework is particularly adept at processing the high-frequency temporal dynamics of IMU signals, seamlessly fusing them with visual and tactile modalities to generate highly stable whole-body locomotion and manipulation actions.

These mobile tasks provide an evaluation of robustness to platform-induced dynamic disturbances, including acceleration, vibration, wheel slip, and odometry drift. They should, however, be distinguished from fully unconstrained dynamic environments containing independently moving external objects or agents.

F. Discussion of Failure Cases

While the FEWT framework significantly improves overall task success rates, analyzing the remaining failures provides critical insights into the system's boundaries. Based on the empirical evaluations, particularly the failures observed in the real-world *Drawer Storage* and *Doll Storage* tasks, we categorize the primary failure modes into the following taxonomy (visualized in Fig. S4):

Drawer Storage

Failed Case (1): The object hit the drawer edge and bounced out during placement.



Failed Case (2): The left arm slipped from the handle and repeatedly tugged, failing to open the drawer.



Doll Storage

Failed Case (1): After release, the doll hit the storage box edge and bounced out.



Failed Case (2): The doll collided with the storage box during placement, causing failure.



Fig. S4: Failure cases of the *Drawer Storage* and *Doll Storage* tasks in real-world experiments.

Placement Misalignment and Edge Collisions:

In long-horizon manipulation, minor spatial deviations can accumulate. During the final placement phase, these micro-errors occasionally cause the manipulated object to misalign with the precise 3D boundaries of the target receptacle. As seen in *Drawer Storage* Failed Case (1) and *Doll Storage* Failed Cases (1) & (2), the object or the doll collides with the rigid edges of the drawer or the storage box. The resulting physical impact forces the object to bounce out or fail to enter the container. This highlights the inherent challenge of achieving millimeter-level spatial precision at the terminal stage of a long-horizon visuomotor policy.

Grasping Slip and Compounding Execution Errors:

Physical interactions in the real world involve complex contact dynamics. As demonstrated in *Drawer Storage* Failed Case (2), the robot's left arm occasionally slips from the drawer handle during the pulling motion due to insufficient grip stability or slight angle deviations. Because our imitation learning policy is trained primarily on optimal expert demonstrations, it lacks explicit data on how to recover from such failures. Consequently, the policy enters an out-of-distribution state, leading to compounding errors where the arm repeatedly tugs without successfully reopening the drawer.

The other potential causes of failure in real-world tasks are summarized as follows:

Mobile Base Odometry Drift:

The wheeled bimanual platform is naturally susceptible to minor wheel slippage and odometry drift during extended navigation phases. In precision-demanding tasks like *Drawer Storage*, even a minor positioning error (e.g., 2-3 cm) of the mobile chassis can shift the target handle out of the robotic arm's optimal kinematic workspace, resulting in a grasping execution failure.

Severe Visual Occlusion:

In tasks involving large or highly deformable objects, such as the *Doll Storage* task, the object itself or the intersecting bimanual arms can severely occlude the visual field of the top-mounted and wrist cameras. When critical geometric features are completely blocked, the visual encoder temporarily loses spatial context, causing the policy to generate incorrect intermediate action chunks and deviate from the target transfer trajectory.

Demonstration Data Bias:

Imitation learning policies are inherently bounded by the distribution of the training dataset. Our current data collection strategy via exoskeleton-style teleoperation primarily captured optimal, unperturbed task executions. Consequently, when the robot encounters unrepresented edge cases during autonomous inference (e.g., approaching the drawer at a suboptimal angle due to base misalignment), the policy lacks the necessary "recovery-from-error" demonstration data to dynamically correct its posture, often leading to compounding errors and task termination.

Addressing these identified failure modes provides a clear roadmap for our future work, specifically pointing toward the need for collecting active "recovery-from-error" demonstrations and further enhancing high-precision 3D spatial perception.

It should be emphasized that the occlusions analyzed above arose naturally during autonomous task execution rather than through a separately controlled occlusion benchmark. These cases therefore serve to identify the practical failure boundary of the current system, rather than to establish invariance to arbitrary occlusion patterns. Moderate and transient occlusions may be compensated for by the complementary camera views, proprioceptive states, IMU feedback, and tactile signals. However, when severe occlusion simultaneously removes task-critical information from all visual views, the current end-to-end policy may enter an out-of-distribution state and generate compounding errors. Future work will investigate explicit occlusion augmentation, active viewpoint adjustment, multi-view 3D perception, and recovery-from-error demonstrations.

G. Sensitivity Analysis and Noise Robustness

To further validate the stability of the Frequency-Enhanced Wavelet-based Transformer (FEWT), we conducted extended sensitivity and robustness analyses. Regarding internal hyperparameter sensitivity, the dynamic weight α within the FE-EMA module is a critical parameter controlling the fusion ratio of cross-spatial and frequency-domain features. As visualized in **Fig. S2**, across both the *Cube Transfer* and *Bimanual Insertion* simulation tasks, the α value rapidly converges and consistently fluctuates within a highly stable range (approximately 0.46 to 0.52) over 2000 epochs. This stable convergence indicates that the model is not overly sensitive to minor hyperparameter initialization variations and successfully maintains an optimal equilibrium between spatial and frequency feature extraction.

In the FEWT framework, the action chunk size (k) dictates the temporal horizon of the predicted action sequence at each inference step. It is the most critical hyperparameter balancing compounding errors and closed-loop reactivity. Following the

established paradigm of the ACT baseline, we set $k=100$ and conducted a sensitivity analysis to validate its optimality within our multimodal framework.

As illustrated in the sensitivity curve (**Fig. S5**), we evaluated the success rate of the highly constrained *Bimanual Insertion* task across various chunk sizes ($k \in \{10, 50, 100, 200, 400\}$).

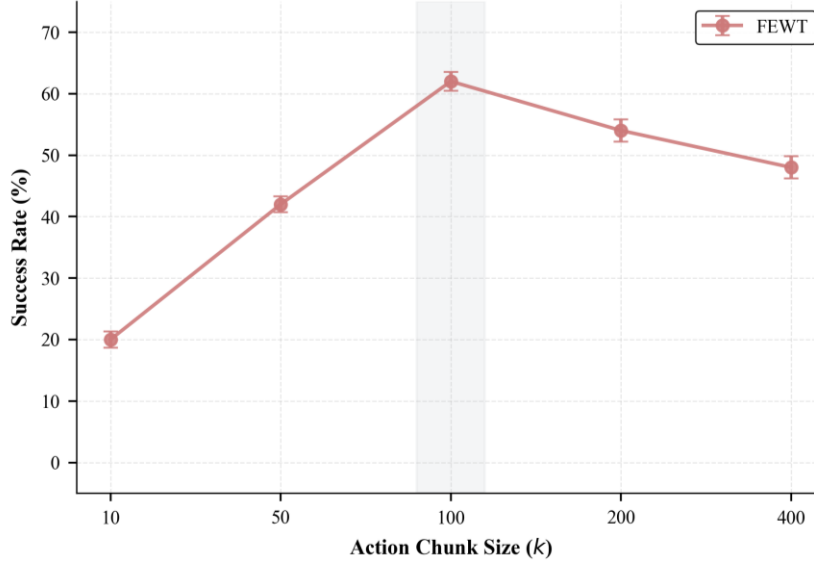


Fig. S5: Sensitivity curve of the chunk size hyperparameter, with different values of chunk size k on the x-axis, and success rate on the y-axis.

The results demonstrate an inverted U-shape performance trend. When k is too small (e.g., $k=10$), the policy effectively degenerates into a near single-step prediction model, making it highly susceptible to compounding errors and non-Markovian pauses in the expert demonstrations. Conversely, as k increases to 200 and 400, the performance experiences a notable decline. This performance dip occurs because excessively large action chunks force the model into a near open-loop execution mode. Therefore, $k=100$ serves as the optimal temporal window, ensuring smooth trajectory generation while preserving sufficient closed-loop reactive capabilities.

Regarding external noise robustness, the inherent mathematical properties of the Time-Series Discrete Wavelet Transform (TS-DWT) provide a structural advantage. In complex, contact-rich manipulation, external dynamic perturbations often manifest as high-frequency sensory noise. By explicitly decomposing features into low-frequency approximations and high-frequency details, the TS-DWT acts as an implicit regularizer. It isolates task-critical high-frequency signals (e.g., genuine rigid impacts) from random environmental high-frequency noise, ensuring that the Transformer Decoder generates robust action chunks even under complex real-world perturbations.

As presented in **Fig. S6**, a quantitative noise robustness evaluation was conducted to assess the stability of the ACT baseline and our proposed FEWT framework under dynamic kinematic perturbations. Uniformly distributed spatial noise, ranging from 0 to an extreme 0.015 m, was injected into the end-effector trajectories during both the relatively simpler *Cube Transfer* and the highly constrained *Bimanual Insertion* simulation tasks.

The empirical results reveal a stark contrast in degradation behavior. As the magnitude of spatial noise increases, the conventional spatial-focused ACT baseline experiences a severe decline. Specifically, in the highly constrained *Bimanual Insertion* task, ACT’s success rate deteriorates sharply to approximately 10% under the maximum 0.015 m noise condition, demonstrating a critical vulnerability to high-frequency spatial jitter.

In contrast, the FEWT framework exhibits a significantly flatter degradation curve and remarkable resilience. Despite the severe 0.015 m perturbations, FEWT maintains a robust success rate of over 50% in the insertion task and over 90% in the transfer task. This substantial performance margin confirms the underlying architectural advantage: the Time-Series Discrete Wavelet Transform (TS-DWT) effectively functions as a multiresolution frequency filter. By explicitly decoupling feature dimensions, TS-DWT successfully dampens redundant high-frequency background noise while preserving the task-critical manipulation signals necessary for stable bimanual coordination.

Taken together, **Figs. S5** and **S6** characterize two complementary dimensions of robustness. **Fig. S5** evaluates temporal sensitivity over action-prediction horizons ranging from 0.2 to 8.0 s, whereas **Fig. S6** evaluates robustness to trajectory-level spatial perturbations up to 0.015 m. These results demonstrate that FEWT maintains improved stability over ACT across the evaluated temporal and kinematic disturbance ranges. Nevertheless, the experiments do not constitute validation under arbitrary

illumination shifts, independently moving external agents, or entirely unseen semantic objects. Such open-world conditions require additional data diversity and training mechanisms beyond the present task-specific imitation-learning setting.

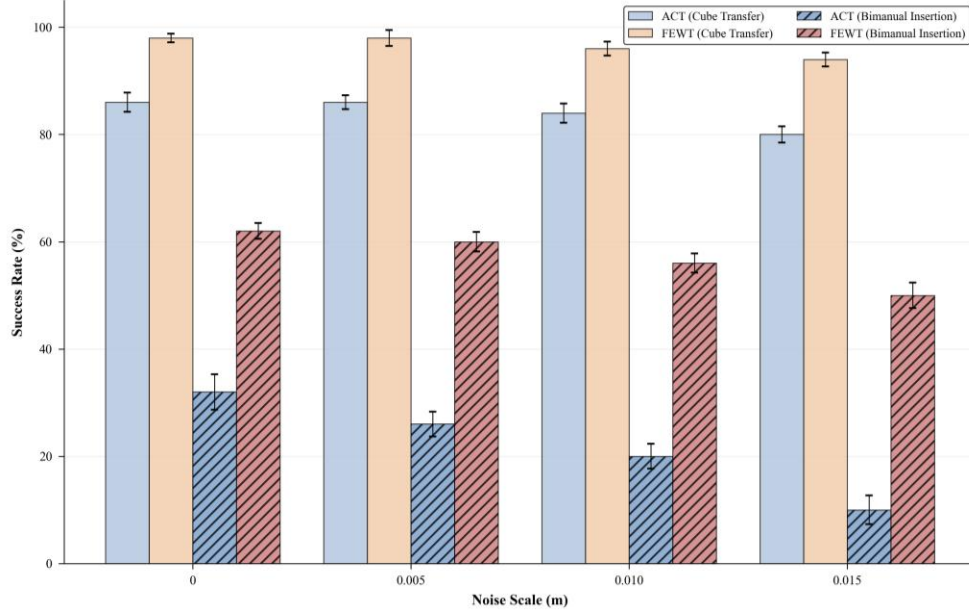


Fig. S6: Quantitative noise robustness evaluation.

H. Computational Efficiency and Scalability Analysis

To provide a more comprehensive evaluation of computational efficiency, we analyze the proposed FEWT framework from five perspectives: module-level computational overhead, practical deployment cost, temporal-horizon sensitivity, and architectural scalability.

Module-level computational overhead

As summarized in **Tables S1** and **S2**, FE-EMA introduces only marginal additional computation under both ACT and Diffusion Policy architectures. Under the ACT–ResNet18 configuration, the parameter count increases from 60.7566 M to 60.7621 M, corresponding to an increase of only 0.0091%. The computational cost increases from 37.57051 G to 37.59691 G FLOPs per sample, representing an overhead of only 0.0703%.

Compared with conventional EMA, FE-EMA contains slightly more profiled parameters but requires 19.68 M fewer FLOPs, corresponding to a reduction of approximately 0.0523% in total forward-pass computation. This behavior arises because the wavelet decomposition separates and down samples the multi-frequency feature components before the principal convolutional enhancement operations are performed. Consequently, the additional frequency-fusion parameters do not lead to a proportional increase in computational cost. Under the substantially more computationally demanding DDIM-based Diffusion Policy, FE-EMA increases the parameter count by only 0.0054% and the forward-pass FLOPs by only 0.1167% relative to the baseline. Conventional EMA introduces 691.25 M additional FLOPs over DDIM, whereas FE-EMA introduces only 396.06 M additional FLOPs. Therefore, FE-EMA reduces the incremental computational overhead introduced by the attention module by approximately 42.70%. These results demonstrate that the lightweight property of FE-EMA remains stable across policy architectures with markedly different model sizes and computational demands.

Performance–computation trade-off

The computational results in **Table S1** were further considered together with the ablation results reported in the main manuscript. With only 0.0091% additional profiled parameters and 0.0703% additional FLOPs over ACT, FE-EMA improves the final Transfer-stage success rate of Cube Transfer from 86% to 100% and the Insert-stage success rate of Bimanual Insertion from 32% to 58%.

Compared with conventional EMA, FE-EMA improves the Insert-stage success rate by 18 percentage points while requiring fewer FLOPs. Therefore, the performance improvement cannot be attributed to brute-force model expansion. Instead, FE-EMA

achieves a favorable performance–computation trade-off by improving the quality of spatial-frequency feature representations, particularly during contact-rich and precision-demanding manipulation phases.

Practical deployment cost

The complete FEWT policy was deployed on a desktop computer equipped with an NVIDIA RTX 3060 GPU. The measured GPU-memory consumption was approximately 4–5 GB during training and approximately 1–2 GB during real-time inference.

The measured end-to-end sensing-to-actuation latency was approximately 60–80 ms. This measurement includes camera-frame buffering, multimodal timestamp synchronization, ROS topic transmission, command publication, and actuator response. Therefore, it represents system-level latency rather than isolated neural-network forward-pass latency.

The low-level controller executes buffered action commands at 50 Hz, whereas the policy generates temporally continuous action chunks and operates asynchronously with the low-level control loop. Consequently, the 50-Hz action-execution rate and the 60–80-ms end-to-end policy-update latency describe different levels of the control pipeline and should not be interpreted as equivalent measurements.

Temporal-horizon scalability

The current FEWT implementation receives a synchronized multimodal observation at each policy query and predicts an action sequence of length k . Therefore, k represents the action-prediction horizon rather than the length of a stacked historical-observation sequence.

At the 50-Hz action-execution frequency, the evaluated values

$$k \in \{10, 50, 100, 200, 400\}$$

correspond to nominal action-prediction horizons of 0.2, 1.0, 2.0, 4.0, and 8.0 s, respectively. This analysis covers a 40-fold range of temporal scales.

As shown in **Fig. S5**, the success rate of *Bimanual Insertion* increases from approximately 20% at $k = 10$ to approximately 62% at $k = 100$, and subsequently decreases to approximately 54% and 48% at $k = 200$ and $k = 400$, respectively. Excessively short action chunks increase sensitivity to local prediction fluctuations and accumulated short-step errors, whereas excessively long action chunks reduce closed-loop responsiveness and approach open-loop execution. Therefore, $k = 100$, corresponding to a nominal 2.0-s prediction horizon, provides the most favorable balance between trajectory continuity, error suppression, and reactive control.

Scalability with Transformer size

Let L denote the encoder-token length, k the action-query length, d the hidden dimension, f the feed-forward dimension, and E and D the numbers of Transformer encoder and decoder layers, respectively. The dominant computational complexities can be approximately expressed as

$$O(E(L^2d + Ldf))$$

for the Transformer Encoder and

$$O(D(k^2d + kLd + kdf))$$

for the Transformer Decoder.

Increasing the numbers of encoder and decoder layers therefore produces approximately linear growth with network depth. Increasing the action horizon primarily affects decoder self-attention and cross-attention, whereas increasing the encoder-token length primarily affects encoder self-attention. If a historical observation window of length h were introduced, the encoder-token length would increase approximately proportionally to h , causing the self-attention term to grow approximately quadratically with the observation-history length.

FE-EMA is applied once to the visual backbone feature maps, while TS-DWT is inserted once between the Transformer Encoder and Decoder rather than being repeated within each Transformer layer. The fixed Haar wavelet decomposition scales approximately linearly with the number of processed feature elements. Therefore, the proposed frequency-enhancement modules do not alter the dominant asymptotic scaling behavior of the Transformer architecture.

The adopted configuration consists of four encoder layers, seven decoder layers, a hidden dimension of 512, a feed-forward dimension of 3200, eight attention heads, and an action-chunk length of $k = 100$. This configuration was selected to balance manipulation performance, closed-loop responsiveness, and practical computational cost.

Network-only wall-clock latency comparisons across all baseline policies and empirical profiling of substantially deeper or wider Transformer variants were not conducted in the present study. These evaluations, together with policies using longer historical observation sequences, will be investigated in future work.

I. Cross-Component Interaction and Task-Dependent Complementarity

The FEWT framework integrates its components through a hierarchical information-processing pathway. FE-EMA first enhances the visual feature maps by combining spatial attention with multiresolution frequency information. The enhanced visual features are subsequently projected into tokens together with joint-state, IMU, and tactile embeddings. After cross-modal interaction in the Transformer Encoder, TS-DWT performs frequency recalibration on the fused memory before action decoding. Therefore, FE-EMA determines the quality and frequency consistency of the visual representation, IMU and STF provide complementary physical observations, and TS-DWT operates on their jointly encoded representation rather than on any raw modality in isolation.

To quantitatively characterize these interactions using the existing controlled ablations, we analyze the conditional marginal contribution of each component. Let

$$\Delta_{\text{TS|A}} = S(A + \text{TS-DWT}) - S(A),$$

where S denotes the task success rate and A denotes the spatial attention mechanism.

During the Insert phase (**Table 2** and **Table S7**), the marginal contribution of TS-DWT under conventional EMA is

$$\Delta_{\text{TS|EMA}} = 36\% - 40\% = -4\%.$$

Under FE-EMA, the corresponding marginal contribution is

$$\Delta_{\text{TS|FE-EMA}} = 62\% - 58\% = +4\%.$$

The resulting compatibility contrast is therefore

$$\Delta_{\text{TS|FE-EMA}} - \Delta_{\text{TS|EMA}} = 8 \text{ percentage points.}$$

This result shows that frequency enhancement within the spatial encoder changes the contribution of TS-DWT from negative to positive in the most constrained contact phase. However, this interaction is task dependent. In the smoother Lifted and Transfer phases, adding TS-DWT to FE-EMA produces a small decrease of 2 percentage points, indicating partial redundancy when abrupt contact dynamics are limited. Thus, the results support improved representational compatibility in contact-rich phases rather than universal super-additive synergy.

Table S7. Staged cross-component complementarity analysis based on controlled ablations.

Controlled comparison	Task	Without target component	With target component	Marginal change
EMA $\rightarrow$ EMA + TS-DWT	Insert (sim)	40%	36%	−4 pp
FE-EMA $\rightarrow$ FE-EMA + TS-DWT	Insert (sim)	58%	62%	+4 pp
Compatibility contrast	Insert (sim)	—	—	+8 pp
FEWT $\rightarrow$ FEWT + STF, IMU fixed	<i>Object Storage</i> (Mobile)	94%	98%	+4 pp
FEWT $\rightarrow$ FEWT + STF, IMU fixed	<i>Bimanual Carry</i> (Mobile)	92%	98%	+6 pp
FEWT $\rightarrow$ FEWT + STF, IMU fixed	<i>Bimanual Transfer</i> (Static)	96%	100%	+4 pp
FEWT $\rightarrow$ FEWT + STF, IMU fixed	<i>Bimanual Insertion</i> (Static)	76%	82%	+6 pp
FEWT + STF, IMU varied	<i>Object Storage</i> (Mobile)	90%	98%	+8 pp
FEWT + STF, IMU varied	<i>Bimanual Carry</i> (Mobile)	86%	98%	+12 pp

* “pp” denotes percentage points. Tactile effects are calculated with IMU feedback retained, whereas IMU effects are calculated with STF retained. The compatibility contrast represents the difference between the marginal effect of TS-DWT under FE-EMA and that under conventional EMA.

The physical modalities exhibit a complementary pattern. With IMU feedback fixed, adding STF to FEWT improves *Object Storage*, *Bimanual Carry*, *Bimanual Transfer*, and *Bimanual Insertion* by 4, 6, 4, and 6 percentage points, respectively, yielding an overall improvement of 4.8 percentage points. With STF fixed, adding IMU feedback improves *Object Storage* and *Bimanual Carry* by 8 and 12 percentage points, respectively, yielding an overall improvement of 10.1 percentage points.

These results reveal a scale-dependent division of information. FE-EMA preserves spatial structure and frequency-aware visual details; IMU feedback captures global platform-level motion; STF captures local contact-state variations; and TS-DWT recalibrates the fused multimodal memory before action decoding. The larger IMU gain in *Bimanual Carry* and the tactile gains in *Bimanual Carry* and *Bimanual Insertion* indicate that the contribution of each modality becomes more pronounced when the corresponding physical uncertainty—global motion or local contact—becomes dominant.

Supplementary **Fig. S6** provides an additional disturbance-level analysis of the integrated FE-EMA–TS-DWT architecture. As the injected spatial perturbation increases to 0.015 m, FEWT exhibits a substantially flatter degradation trend than ACT, with the performance separation becoming most pronounced in the more constrained *Bimanual Insertion* task. This result is consistent with the task-dependent complementarity observed in **Table 2**: the benefit of frequency-aware spatial and temporal representations increases when precise alignment and rapid correction are simultaneously required. The present experiments employ staged controlled ablations: simulation isolates the FE-EMA–TS-DWT interaction, **Table 5** evaluates tactile sensing with IMU retained, and **Table S6** evaluates IMU feedback with STF retained. We therefore describe the observed relationship as task-dependent complementarity rather than statistically exhaustive higher-order synergy.